This is going to be a loosely held-together collection of notes on different types of constructions in the field of operator algebras whose primary connection is that they're referred to by "product."

Tensor Products of C^*-Algebras

One of the main complications with C^*-algebras is that the algebraic tensor product $A \odot B$ may admit more than one C^*-norm.

Definition: A C^*-norm on $A \odot B$ is a norm $\|\cdot\|_\alpha$ satisfying $\|xy\|_\alpha \leq \|x\|_\alpha \|y\|_\alpha$ and $\|x^*x\|_\alpha = \|x\|_\alpha^2$ for all $x, y \in A \odot B$. The completion of $A \odot B$ with respect to this norm is denoted $A \otimes_\alpha B$.

However, the existence of there being multiple norms is not necessarily universal.

For instance, since there is a $*$-isomorphism between $\mathbb{M}_n(\mathbb{C}) \odot A$ and $\mathbb{M}_n(A)$, we have that $\mathbb{M}_n(\mathbb{C}) \odot A$ admits a unique C^*-norm, which is the norm that it obtains from $\mathbb{M}_n(A)$.

Definition: Let A and B are C^*-algebras. The *maximal norm* on $A \odot B$ is given by

$$\|x\|_{\max} = \sup\{\|\pi(x)\| \mid \pi\colon A \odot B \to B(H) \text{ a cyclic } *\text{-representation}\}$$

for any $x \in A \odot B$. The completion of $A \odot B$ with respect to $\|\cdot\|_{\max}$ is denoted $A \otimes_{\max} B$.

This is akin to the universal C^*-norm on the group C^*-algebra $C^*(\Gamma) \subseteq B(\ell_2(\Gamma))$.

Definition: If $\pi\colon A \to B(H)$ and $\sigma\colon B \to B(K)$ are faithful representations, then the *spatial* or *minimal* C^*-norm on $A \odot B$ is given by

$$\left\|\sum_{i=1}^{n} a_i \otimes b_i\right\|_{\min} = \left\|\sum_{i=1}^{n} \pi(a_i) \otimes \sigma(b_i)\right\|_{B(H \otimes K)}.$$

The completion of $A \odot B$ with respect to $\|\cdot\|_{\min}$ is denoted $A \otimes_{\min} B$.

Remark: If we consider von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, then there are numerous C^*-norms that $M \odot N$ can be endowed with, but their norm completions will not be von Neumann algebras. The von Neumann algebra tensor product is still well-defined, and denoted $M \bar{\otimes} N$, which is the von Neumann algebra generated by $M \otimes N \subseteq B(H \otimes K)$.

There are a variety of technicalities we must consider before moving forward. First, we must verify that $\|\cdot\|_{\max}$ is finite; for this, observe that if $\pi\colon A \odot B \to B(H)$ is any representation, then we may restrict to π_A and π_B, such that

$$\begin{aligned}\|\pi(a \otimes b)\| &\leq \|\pi_A(a)\|\|\pi_B(b)\| \\ &\leq \|a\|\|b\|,\end{aligned}$$

meaning $\|x\|_{\max} < \infty$ for all $x \in A \odot B$.

The other major questions we must consider are the following.

(i) Are $\|\cdot\|_{\max}$ and $\|\cdot\|_{\min}$ even C^*-norms?

(ii) Is $\|\cdot\|_{\min}$ independent of the choice of faithful representation?

(iii) Is it possible to reduce the non-unital case to the unital case?

We can dispense with (iii) by observing that, in fact, any C^*-norm on the tensor product $A_1 \odot A_2$ can be uniquely extended to the tensor product of their unitizations $\widetilde{A_1} \odot \widetilde{A_2}$.

Proposition: Let A_1 and A_2 be non-unital C^*-algebras with respective unitizations $\widetilde{A_1}$ and $\widetilde{A_2}$. Then, if $\|\cdot\|_\beta$ is any C^*-norm on the tensor product $A_1 \odot A_2$, it can be extended to a C^*-norm on $\widetilde{A_1} \odot \widetilde{A_2}$.

Proof. Denote by $A_\beta = A_1 \otimes_\beta A_2$. Let $\pi\colon A_\beta \to B(H)$ be a faithful representation, meaning that $\|x\|_\beta = \|\pi(x)\|$ for all $x \in A_1 \odot A_2$.

If π_1 and π_2 are the respective restrictions of π to A_1 and A_2, then we may extend π_1 and π_2 to the uniti-

zations $\widetilde{A_1}$ and $\widetilde{A_2}$ respectively by taking

$$\pi_i^0(x + \lambda 1) = \pi_i(x) + \lambda 1$$

for each $x \in A_i$ and $\lambda \in \mathbb{C}$. The ranges of π_1^0 and π_2^0 commute, meaning that we may define a representation π_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by

$$\pi_0(x_1 \otimes x_2) = \pi_1^0(x_1)\pi_2^0(x_2).$$

Observe that π_0 extends π. This allows us to define a C^*-norm β_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by taking $\|x\|_{\beta_0} = \|\pi_0(x)\|$. This is our desired extension to the unitization. □

Tackling questions (i) and (ii) is more difficult. For this, consider the fact that $\|\cdot\|_{\max}$ admits a universal property as described below, which follows from the fact that any C^*-algebra can be faithfully represented on some Hilbert space.

Proposition: Let $\pi\colon A \odot B \to C$ be a $*$-homomorphism. Then, there exists a unique $*$-homomorphism $A \otimes_{\max} B \to C$ that extends π. In particular, a pair of $*$-homomorphisms $\pi_A\colon A \to C$ and $\pi_B\colon B \to C$ with commuting ranges induces a unique $*$-homomorphism on $A \otimes_{\max} B$.

Free Products and Ultraproducts

Crossed Products

If we have a group Γ acting on a C^*-algebra A, then a question becomes how we may construct a C^*-algebra that encodes this action.

If a group is acting on another group, there is a construction known as the (outer) semidirect product that enables this. We will construct a C^*-algebra emerging from an action $\alpha\colon \Gamma \to \operatorname{aut}(A)$, which will be dubbed $A \rtimes_\alpha \Gamma$.